PHIL 133 · MAKING MINDS · FALL 2026

Midterm Prep

Thursday, November 5 · 12:30–1:45pm · in class

To prepare for the exam, make sure that you can solve the following problems with just pencil and paper.

+1 B Any circuit you solve can be boxed and used later.

Combinatorial Circuits CCs

BASIC CIRCUITS

- ☐ NAND
- ☐ XOR
- ☐ Always 1
- ☐ Half adder

TALLY

- ☐ +1 [0–4] T
- ☐ +2 [0–4] T
- ☐ +3 [0–4] T

BINARY

- ☐ +1 [0–15] B
- ☐ +2 [0–15] B
- ☐ +3 [0–15] B
- ☐ $2x$ [0–15] B
- ☐ $4x$ [0–15] B
- ☐ $x + y$ [0–15] B

SIMPLE COMBINATIONS

- ☐ $2x + 1$ [0–15] B
- ☐ $2(x + 1)$ [0–15] B

Sequential Circuits SCs

BASIC CIRCUITS

- ☐ Trip-wire
- ☐ ON/OFF switch

TALLY

- ☐ +1 T
- ☐ +2 T
- ☐ +3 T

BINARY

- ☐ +1 B
- ☐ +2 B
- ☐ +3 B
- ☐ $2x$ B
- ☐ $4x$ B
- ☐ $x + y$ B

COMBINATIONS: ONE VARIABLE

- ☐ $2x + 1$ B
- ☐ $2(x + 1)$ B
- ☐ $3x$ B
- ☐ $3x + 2$ B
- ☐ $4(x + 2)$ B
- ☐ $3(2x + 2)$ B

COMBINATIONS: TWO VARIABLES

- ☐ $x + y + 1$ B
- ☐ $2x + y$ B
- ☐ $2(x + y)$ B
- ☐ $3x + 2y$ B

NAVIGATION

- ☐ Zig-zag
- ☐ 4-forward/4-back

POSSIBLE WITH AN SC?

Design the circuit, or explain why no SC can do it.

- ☐ $x \cdot y$ B
- ☐ $2x$ T
- ☐ $x + y$ T
- ☐ Mad Max